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From Narrative Patterns to Institutional Design: A Lua-Based Framework for Identifying and Mitigating Societal Collapse Invariants

Foundational Invariant-Based Analytical Framework

The LiteratureHero project is built upon a sophisticated analytical framework designed to systematically deconstruct both fictional and real-world texts to identify recurring structural patterns associated with societal collapse. This methodology transcends simple plot summarization by focusing on a curated set of over twenty "invariants," which are defined as persistent, cross-cultural patterns of thought, behavior, and social organization that frequently precede or characterize societal breakdown . By treating these invariants as analytical lenses, the system can move beyond surface-level narrative elements to uncover deeper, more universal drivers of conflict and systemic failure. The core innovation lies in its ability to map fictional tropes directly onto documented historical phenomena, creating a robust bridge between speculative fiction and empirical reality. For instance, the "Unity / Hive Mind" invariant connects the fictional concept of a collective consciousness to the historical reality of totalitarian collectivism, using specific case studies like the Maoist Red Guards during China's Cultural Revolution as a real-world anchor . Similarly, the "Tech Gatekeeping" invariant links the trope of a technocratic priesthood controlling critical infrastructure to the actions of high-stakes doomsday cults such as Japan's Aum Shinrikyo, which recruited elite scientists to build secret weapons programs [42]. This mapping provides a rigorous, evidence-based foundation for analyzing narratives, transforming them from entertainment into a source of structured data about potential societal vulnerabilities.

The application of this invariant-based lens reveals that many common post-apocalyptic narrative tropes are not merely dramatic devices but plausible outcomes of specific social and psychological pressures. The case study of the Mycelium cult from *Atom RPG* serves as a prime example of this analytical power . On the surface, Mycelium presents itself with benign principles of "Unity," "Love," "Modesty," and "Life," which closely mirror the functions of charitable religious organizations that often emerge after great wars . However, a deeper analysis through the invariant framework uncovers a far more

dangerous underlying structure. The principle of "Unity" is revealed not as mere solidarity but as a demand for total personality dissolution into a collective consciousness, a direct manifestation of the Unity/Hive Mind invariant . The cult's "Love" and "Modesty" are shown to be part of a "Utopian Principles Weaponized" strategy, where genuine acts of charity serve as a recruitment tool and a moral shield for increasingly coercive control measures . The system highlights how the combination of these seemingly positive traits —charity, unity, and a focus on technology stewardship—creates a composite profile of a post-war, science-flavored new religious movement with high-control and technocratic gatekeeping characteristics . This reframes why such groups are perceived as antagonists in fiction: their methods, even when aimed at a "greater good," involve stripping away the individual autonomy necessary for a pluralistic and resilient recovery .

This analytical approach allows LiteratureHero to function as a "stress test" for social systems, simulating how various combinations of invariants interact under the extreme duress of collapse. The framework demonstrates that it is rarely a single factor but the convergence of multiple invariants that leads to catastrophic outcomes. For example, the presence of "Doomsday Prophecy" combined with "Sacred Violence" can lead to apocalyptic scenarios where mass violence is framed as a necessary purification ritual . When these patterns are identified, the system moves beyond diagnosis to prevention. It translates the observed dynamics into concrete, actionable insights. The analysis of Mycelium suggests real-world design lessons, such as the need for independent oversight of critical technology stewards, the establishment of secular accountability structures for movements that provide social services, and the creation of pathways to community and meaning that do not require total fusion into a hive mind . By quantifying and comparing the strength of different invariants across various texts—from video games like *Fallout* to real-world policy documents—the system can rank narratives and even real-world discourse samples by their proximity to dangerous collapse trajectories [137]. This quantitative comparison provides a powerful tool for identifying realistic failure modes and drafting counter-measures, or "escape routes," for policymakers and civil society . The ultimate goal is to separate fact from fiction, identify plausible collapse pathways, and generate the institutional designs needed to mitigate the risks they pose .

Invariant Name	Fictional Trait Example	Historical Real-World Pattern
Unity / Hive Mind	Individuals dissolve into a unified collective consciousness; dissent is treason .	Totalitarian collectivism demanding fusion of self and state/party, enforced through mutual surveillance and struggle sessions .
Tech Gatekeeping	A small technocratic group controls access to critical infrastructure like reactors or AIs .	Closed military research complexes, nuclear elites, and regimes guarding technical knowledge as a state secret to maintain power .
Utopian Principles Weaponized	Movements begin with egalitarian goals that justify increasing absolutist control "for the greater good" .	Revolutionary regimes and communal experiments that slide from reform to repression, normalizing purges and show trials as "necessary sacrifices" .
Doomsday Prophecy	An impending cataclysm is treated as inevitable, justifying preemptive violence .	Millennialist and apocalyptic movements that see collapse as divinely scripted, framing mass action as the only path to salvation .
Charismatic Prophetic Leader	A single leader is seen as the sole interpreter of fate or higher intelligence, whose authority collapses the group's logic .	Personality cults around political or religious leaders elevated as semi-divine figures, where succession crises trigger splinter wars .
Sacred Violence & Purification	Killing enemies or "unclean" populations is framed as a ritual act of purification to restore the world .	Ethnic cleansing, inquisitions, and ideological "cleansing" campaigns where quasi-religious language overrides moral inhibitions against mass killing .

Technical Architecture: A Dependency-Free Lua Ecosystem

The technical implementation of LiteratureHero is centered on a lightweight, modular, and dependency-free ecosystem built entirely within the Lua programming language. This architectural choice is a deliberate strategic decision to ensure portability, transparency, and ease of modification, aligning with the project's goal of creating a professional, GitHub-ready, and self-contained framework . Lua is recognized for its efficiency and suitability for embedding in larger applications, making it an ideal candidate for a scripting engine that needs to operate without reliance on extensive external services or libraries [13] [17]. The proposed system architecture is meticulously organized into distinct, decoupled modules, each with a singular responsibility, which collectively form a cohesive and scalable analysis pipeline . At the heart of this structure is the `lh_engine.lua` module, which serves as the central orchestrator of the entire analysis process . It is designed to be a thin layer of control that accepts a specification for a work to be analyzed—a simple table containing metadata and file paths—and then sequentially calls upon other modules to perform loading, processing, and analysis tasks . This separation of concerns ensures that the core logic of the engine remains clean and focused on coordination rather than complex rule implementation.

The system's modularity is further reinforced by dedicated modules for input, processing, and output. The `text_loader.lua` module is responsible for ingesting raw text from various sources, such as game scripts, wiki pages, or real-world document corpora, and normalizing it into a consistent internal format of tables containing the text content and associated metadata . This decouples the analysis logic from the specifics of file formats and encodings, allowing the system to be easily adapted to new types of source material without altering the analytical modules . Following ingestion, the `tokenizer.lua` module performs basic linguistic preprocessing, breaking down the normalized text into tokens (words) and sentences . This foundational step provides the necessary structure for all subsequent invariant detection modules to operate effectively, without resorting to heavy external Natural Language Processing (NLP) libraries, which is a key constraint of the project . The architecture scales gracefully through a registry pattern managed by the `src/invariants/invariant_registry.lua` file . This registry acts as a master index, mapping the string names of invariants (e.g., `"unity_hivemind"`) to their corresponding check functions or modules . This design makes the system highly extensible; adding a new invariant simply involves creating a new module and registering it, without requiring any changes to the core engine or other existing modules . This table-driven approach leverages one of Lua's core strengths: its flexible use of tables for data storage and lookup [26].

The final components of the architecture are designed to transform the raw analytical results into human- and machine-readable formats. The `report_markdown.lua` module takes the comprehensive analysis result table generated by the engine and formats it into a well-structured Markdown document, complete with sections for invariant scores, faction comparisons, and prevention insights . This directly addresses the project's goal of producing professional, GitHub-ready documentation that can be easily shared and understood . To facilitate integration with other tools and systems, the `src/io/export_json.lua` module provides a parallel serialization function, converting the same analysis result into a standardized JSON format . This creates a bridge between the custom Lua framework and the broader data ecosystem, enabling downstream processes such as data visualization, dashboard creation, or feeding into other machine learning models [33]. Finally, the `src/apps/cli_lh_runner.lua` script provides a command-line interface (CLI), significantly lowering the barrier to entry for users who may not be programmers . It parses user arguments to specify the work to be analyzed and the desired output formats, then executes the full pipeline and saves the resulting reports to disk . This collection of 15 files, including the core engine, invariant modules, and supporting tools, constitutes a minimal yet powerful Lua ecosystem that is fully contained, transparent, and designed for practical utility in the field of crisis prevention . The entire structure is designed to be reproducible and modular, reflecting best practices in scientific computing pipelines [14] [32].

Advanced Internal Linguistic Analysis Without External Dependencies

A central requirement for the LiteratureHero system is the implementation of "advanced internal linguistic analysis" while strictly avoiding all external dependencies, including popular NLP libraries like NLTK or spaCy [30]. This constraint necessitates a creative and pragmatic approach that leverages Lua's capabilities to build sophisticated pattern recognition rules from the ground up. While true semantic understanding, sentiment analysis, and entity linking typically rely on large pre-trained models and significant computational resources [21] [106], the system achieves a high degree of analytical rigor through a combination of custom tokenization, semantic pattern matching via knowledge graphs, and narrative structure parsing. The cornerstone of this approach is a custom-built `tokenizer.lua` module. Instead of relying on off-the-shelf solutions, this module would be designed with Lua's string manipulation functions to perform sentence splitting and word tokenization tailored to the specific needs of invariant detection . More importantly, it could be enhanced to identify multi-word phrases, or lexical bundles, which are often more indicative of an underlying theme than single words [70] [73]. For example, detecting the phrase "merger with the collective consciousness" is a much stronger signal for the "Unity / Hive Mind" invariant than simply finding the words "collective" or "unity" in isolation. This attention to phraseology allows the system to capture more nuanced linguistic cues embedded in the text.

To move beyond simple keyword matching, the system can employ a lightweight, internally constructed knowledge graph to model semantic relationships. The definitions of the 20+ invariants themselves form the basis of this graph, with each invariant acting as a node and its associated keywords, synonyms, and conceptual themes forming the edges connecting to other nodes [26]. When the tokenizer processes a text, it doesn't just flag keywords; it attempts to link detected terms to their corresponding nodes in this knowledge graph. For instance, upon encountering the term "purification" in a text discussing the elimination of a certain group, the system would traverse the graph to find connections to the "Sacred Violence" and "Purification" invariants. This method, while simpler than the vector space models used by LLMs, provides a structured way to infer meaning and context based on the predefined analytical vocabulary [153]. This approach is particularly effective for tracking concepts like "resource scarcity," where the system can recognize variations such as "water rationing," "energy crisis," or "arable land shortage" as manifestations of the "Resource-Scarcity Militarization" invariant . The success of this method depends on the thoroughness of the initial knowledge graph construction, which requires deep sociological and literary analysis to define the rich semantic fields associated with each invariant.

Furthermore, the system can incorporate rudimentary narrative structure parsing to understand the context in which linguistic patterns appear. Although building a full-fledged parser is outside the scope of a dependency-free Lua engine, Lua functions can be written to identify common structural elements in texts, especially those derived from interactive media like role-playing games [56]. For example, the system could be programmed to recognize the typical quest structure found in RPGs, distinguishing between dialogue spoken by a non-player character (NPC), objective descriptions provided to the player, and environmental descriptions [56] [57]. This contextual awareness is crucial for accurate analysis. The statement "We must eliminate the unclean to save the future" carries a very different weight if it appears in a quest objective versus a piece of background lore. The system can also track pronouns and shifts in point of view to identify narrative framing, a technique relevant to discourse analysis [46] [49]. For instance, a consistent use of third-person omniscient narration might be linked to the "Ideological Fossilization" invariant, where a single doctrine becomes an unassailable, god-like truth. While these parsing capabilities are limited compared to what is possible with modern AI, they represent a significant step toward "advanced" analysis by moving beyond flat text scanning to engage with the text's structure and rhetorical form [4]. The combination of custom tokenization, a knowledge-graph-based semantic model, and basic narrative parsing allows the LiteratureHero system to meet its ambitious analytical goals within the strict constraints of a self-contained Lua environment.

Hoax Structures and Disinformation as Analytical Invariants

A significant evolution of the LiteratureHero framework is its expansion to include "hoax structures" as a category of analyzable invariants. This development acknowledges that the spread of false information and the formation of conspiratorial belief systems are critical precursors to societal collapse, often working in tandem with other collapse-related invariants . The project uses the infamous Soviet-era television stiob, the "Lenin was a Mushroom" hoax, as a primary case study to deconstruct the rhetorical mechanics of such hoaxes . This analysis yielded a formalizable template for how hidden truth narratives are constructed, which can now be coded into the system as a set of reusable analytical invariants. These invariants allow LiteratureHero to not only detect the presence of a conspiracy or secret history within a fictional work but also to diagnose the specific techniques being used to make it persuasive. This capability is crucial for understanding how disinformation operates and for developing countermeasures.

The analysis of the "Lenin was a Mushroom" hoax identified several key structural invariants that the LiteratureHero system can now formally recognize and measure . One such invariant is the **Authority Mask Invariant**, where the hoax is presented by a character performing as a calm, precise expert—a historian, scientist, or archivist—framing the outlandish claim as a serious investigation rather than satire . Another is the **Chain-of-Evidence Invariant**, where the hoax builds a long sequence of weak individual pieces of "evidence" (ambiguous photos, cherry-picked letters, pseudo-scientific diagrams) whose cumulative effect creates a sense of plausibility despite the weakness of each component . The **Edge-of-Believability Invariant** captures the careful calibration of the hoax's claims to sit precisely on the threshold of what a traumatized or disillusioned audience might accept, exploiting a context of widespread revelation and distrust . Finally, the **Reframe-Founding-Myth Invariant** describes the tactic of reinterpreting the core foundational event of a culture or ideology in a radical new way, often suggesting it was secretly controlled by a hidden force known only to a select few . These invariants provide a clear, rule-based method for the Lua engine to scan a text and flag passages that exhibit the hallmarks of a well-constructed hoax.

By modeling these structures, LiteratureHero gains the ability to analyze the epistemological foundations of belief systems depicted in literature. The system can answer questions like: Is this cult's doctrine built upon a hoax-like narrative? How vulnerable would an in-universe population be to believing this secret history? . This analytical capability is directly applied in the case study of *Atom RPG*'s Mycelium cult. The game's narrative itself incorporates a parody of the Lenin-mushroom hoax, turning a joke about Soviet propaganda into a playable, diegetic cult doctrine . LiteratureHero can now detect this meta-layer, recognizing that Mycelium's seemingly scientific and charitable front is constructed using the very same rhetorical techniques that made the original hoax so effective. This insight transforms the analysis from a simple comparison of traits to a deeper examination of the mechanisms of persuasion. The system can flag the "deadpan expert performance," the "documentary grammar" of faux archives, and the "associative logic" that links visual coincidences to prove a cosmic design—all techniques extracted from the Pyatoe Koleso episode . This allows the project to explore how audiences react to such hoaxes, noting the ambiguous reception where some characters laugh, some half-believe, and some convert, a pattern that reflects how real-world hoaxes create layered belief systems that can be exploited for recruitment . Ultimately, this extension of the invariant framework allows LiteratureHero to model not just the content of a belief system but the very process by which that belief is formed and disseminated, providing a powerful tool for understanding and resisting the forces of disinformation that can destabilize societies.

Generating Actionable Outputs for Crisis Prevention

The LiteratureHero framework is fundamentally oriented towards action, with the explicit goal of translating its deep textual analysis into three distinct, practical outputs designed to prevent societal collapse. These outputs are tailored to three different audiences: policymakers who design institutions, crisis monitors who track risks, and the general public who constitute the body politic. This tripartite approach ensures that the system's findings are not confined to academic discourse but are transformed into usable intelligence for real-world decision-making and public education . Each output type is generated through a specific module in the Lua pipeline, ensuring a systematic and repeatable process for deriving actionable insights from literary and historical data. The generation of these outputs represents the culmination of the entire analytical process, from text ingestion to invariant scoring, and is the key to fulfilling the project's mission of creating an "earth-saving tool" .

The first output type consists of **institutional recommendations** for policymakers and institutional designers. These recommendations are derived directly from the identification of high-risk invariant combinations. When the system detects a dangerous convergence of patterns—such as "Utopian Principles" coupled with "Tech Gatekeeping"—it can generate specific, evidence-based suggestions for mitigating those risks . For example, if the analysis of a contemporary social movement reveals strong indicators of the "Charismatic Prophetic Leader" and "Information Control" invariants, the report could recommend establishing legal and structural safeguards for leadership succession, promoting media pluralism, and creating independent audit trails for information dissemination. These recommendations are grounded in the historical failures of movements like Jim Jones' Peoples Temple, which began with social service but devolved into a closed, paranoid regime under a single leader [68] . The system essentially reverse-engineers the failure modes identified in fiction and history to propose positive-sum institutional designs that promote resilience, such as transparent governance, decentralized technological control, and protected channels for dissent [92] [126]. This output directly supports UNDP frameworks for strengthening institutional and legislative systems and contributes to the global effort to combat corruption and improve integrity [115] [126].

The second output is a set of **early-warning indicators** for crisis monitoring. This function is powered by the system's risk-scoring mechanism, implemented in a dedicated Lua module (`src/tools/risk_scorer.lua`) . This module aggregates the scores from all detected invariants, applying configurable weights from the `lh_config.lua` file to compute composite risk indices, such as "Autonomy Risk," "Doomsday Escalation Risk," or "Technological Capture Risk" . These indices transform qualitative observations

into quantitative metrics that can be tracked over time. For instance, an analyst could use the system to monitor a corpus of political speeches, news articles, and social media posts for spikes in rhetoric that match the "External Scapegoat" and "Ideological Fossilization" invariants. A sustained increase in these scores could serve as an early warning sign of rising authoritarianism or polarization, analogous to the methodologies used by the OECD to monitor multidimensional fragility across economic, political, security, societal, and environmental dimensions [122][137][140]. The system's ability to quantify the presence of these patterns allows it to function as a diagnostic tool for societal health, helping organizations like the UN Development Programme to get ahead of the crisis curve by identifying risks before they escalate [99][114]. The output would be a dashboard or a ranked list of emerging threats, providing timely alerts that enable proactive intervention.

The third and most accessible output is a suite of **public educational materials**. This is the primary function of the `report_markdown.lua` module, which converts the complex analytical results into clear, well-structured, and GitHub-ready Markdown documents . These reports are designed to be understandable by a non-expert audience, serving to debunk myths, explain the psychological appeal of doomsday cults, and illustrate how real-world institutions can resist the pull of totalitarian structures . For example, a report on *Atom RPG*'s Mycelium cult would not only detail the invariant analysis but also summarize the key takeaways for prevention, explaining why combining charity with secrecy and absolute ideology can be dangerous . These materials are intended to foster a more "satire-literate" public that is better equipped to recognize the rhetorical tricks of hoaxes and propaganda . This educational function is vital for building societal resilience, complementing the work of organizations promoting global citizenship education and climate literacy [67][69]. By putting the power of analysis into the hands of the public, LiteratureHero aims to cultivate a more informed and critically aware citizenry, capable of recognizing and rejecting the narratives that lead to societal collapse [146].

System Implementation: The Proposed Lua File Structure and Pipeline

The practical realization of the LiteratureHero framework is encapsulated in a proposed system of 15 interdependent Lua files, organized into a logical directory structure that promotes modularity, maintainability, and usability . This file structure represents a complete blueprint for the system's codebase, providing clear specifications for each

component's purpose, structure, and role in the overall analysis pipeline. The design emphasizes a dependency-free, internal-only approach, leveraging Lua's native features to create a self-contained and portable application . The entire system can be envisioned as a lightweight, table-driven pipeline that orchestrates the flow of data from raw text ingestion to the generation of insightful, actionable reports [14] [37]. This section details the proposed file plan, explaining how each script fits into the larger framework to fulfill the project's ambitious goals.

The core of the system resides in the `src/core/` directory, which contains the fundamental modules for orchestration and configuration. The `lh_engine.lua` file is the central nervous system, exposing a primary API function like `analyze_work(work_spec)` that serves as the single entry point for initiating an analysis . It is responsible for coordinating the entire workflow: loading the specified text files, passing them through the tokenizer, invoking the invariant checkers, running the hoax detector, and finally collating all the results into a unified `analysis_result` table . Separately, `lh_config.lua` provides a centralized location for all tunable parameters, returning a configuration table with values for file paths, risk score thresholds, and the relative weights assigned to each invariant . This decouples the system's behavior from its code, allowing analysts to easily adjust the system's sensitivity or focus without needing to edit the core logic [102]. The `src/io/` directory houses modules for interacting with the external environment. `text_loader.lua` manages the ingestion of data, with functions like `load_text(path)` that read files and return normalized `{id, text, metadata}` tables, ensuring consistency across different input sources . Complementing this, `export_json.lua` provides a mechanism to serialize the final analysis results into a standard JSON format, facilitating integration with other tools, databases, or visualization platforms .

The analytical heart of the system is composed of the modules in `src/invariants/` and `src/analysis/`. The `variant_registry.lua` file acts as a dispatcher, maintaining a table that maps invariant names to their respective check functions, allowing the engine to dynamically run all registered checks . Individual invariant modules like `unity_hivemind.lua`, `tech_gatekeeping.lua`, and `utopian_principles.lua` contain the specific logic for detecting their target pattern . Each exports a `check(doc, context)` function that analyzes a text document and returns a structured result including a numerical score (e.g., from 0 to 1) and excerpts of text that support the finding . To handle more complex analyses, specialized modules exist in the `src/analysis/` directory. The `hoax_detector.lua` module implements the structural invariants derived from the Lenin-mushroom hoax, searching for signs of "Authority Mask," "Chain-of-Evidence," and similar rhetorical patterns . Another key analysis module, `faction_comparator.lua`, is designed to

compare multiple factions within a single work, running invariant checks on each and presenting the results in a comparative table that highlights differences in risk profiles . This automates the kind of analysis seen in the comparison of Mycelium, ATOM Organization, and warlords in *Atom RPG*, making it easy to see which groups pose the highest threat to autonomy and stability .

Finally, the system includes tools and applications that bridge the gap between raw analysis and human consumption. The `risk_scorer.lua` module aggregates the individual invariant scores into higher-level composite risk metrics, providing a concise summary of a text's overall danger profile . The `dialogue_sampler.lua` module works in tandem with the reports, automatically extracting and including short, representative excerpts of text that illustrate each detected invariant, grounding the abstract scores in concrete evidence . The user-facing components are located in `src/apps/`. The `cli_lh_runner.lua` script provides the command-line interface, parsing user inputs and driving the entire analysis process, thereby making the system accessible to non-programmers . Lastly, the `report_markdown.lua` module consumes the final analysis result and generates the primary output: a professional, well-formatted Markdown document ready for GitHub or direct use in reports . This comprehensive 15-file structure provides a complete, end-to-end solution for the LiteratureHero project, from code to actionable insight.

Filename	Directory	Purpose and Structure
`lh_engine.lua`	`src/core/`	Central orchestrator. Exposes `analyze_work()` to load text, call modules, and collate results into a single `analysis_result` table. Acts as the pipeline brain stem .
`lh_config.lua`	`src/core/`	Configuration hub. Returns a table of global constants (paths, thresholds, invariant weights) to make the system behaviorally tunable without code changes .
`text_loader.lua`	`src/io/`	Input handler. Functions like `load_text(path)` normalize ingestion of raw text from various sources into a consistent `{id, text, metadata}` table format .
`tokenizer.lua`	`src/nlp/`	Linguistic preparer. Performs custom word and sentence tokenization using Lua's string library to provide the basic structure needed by invariant modules .
`variant_registry.lua`	`src/invariants/`	Dispatcher. A table mapping invariant names to their check modules, allowing the engine to dynamically run all invariants without hard-coded calls .
`unity_hivemind.lua`	`src/invariants/`	Invariant checker. Exports a `check()` function that searches for lexical cues ("hive," "collective") and structural signals (commands to surrender self) to score "Unity / Hive Mind" patterns .
`tech_gatekeeping.lua`	`src/invariants/`	Invariant checker. Implements a `check()` function to scan for tech-related terms combined with exclusivity language ("chosen," "worthy") to detect "Technocracy vs. Democracy" scenarios .
`utopian_principles.lua`	`src/invariants/`	Invariant checker. Contains functions to look for "love"/"charity" language alongside loyalty demands, returning both a positivity score and a "control overlay" score .
`hoax_detector.lua`	`src/analysis/`	Hoax analyzer. Models "Lenin mushroom" style hoaxes by checking for "Authority Mask," "Chain-of-Evidence," and "Reframe-Founding-Myth" invariants in the text .
`report_markdown.lua`	`src/report/`	Output generator. Accepts an `analysis_result` table and produces a formatted markdown string, creating the primary human-readable, GitHub-ready report .
`faction_comparator.lua`	`src/analysis/`	Comparative tool. Accepts multiple faction profiles, runs invariant checks on each, and returns a side-by-side comparison table of scores and notes .
`risk_scorer.lua`	`src/tools/`	Aggregator. Computes composite risk indices (e.g., "autonomy risk") by applying weights from `lh_config.lua` to the scores of individual invariant checks .
`dialogue_sampler.lua`	`src/tools/`	Evidence extractor. Given an invariant result, selects and returns short, representative text excerpts to provide concrete proof for the analysis in reports .
`export_json.lua`	`src/io/`	Machine-readable exporter. Serializes the `analysis_result` table into a clean JSON document for use in dashboards, visualizations, or other automated systems .
`cli_lh_runner.lua`	`src/apps/`	Command-line interface. A small script that parses arguments (e.g., `--work`, `--input`) and drives the analysis pipeline, lowering the barrier to use for non-programmers .

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